ModelIt!

Teacher's Guide

Why Does Your Brain Lie to You?

Modeling Cognitive Bias, Perception, and Decision-Making in the Brain

Quick Reference

Grade Level	Grade 12L1
NGSS Standard	HS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to predict how cognitive biases affect decision accuracy based on sensory input clarity, prior experience strength, and cognitive load.
Disciplinary Core Idea	**LS1.A Structure and Function** — The nervous system processes sensory information through hierarchical neural networks that use shortcuts (heuristics) to manage the enormous volume of incoming data, producing systematic biases.
Crosscutting Concept	**Cause and Effect** — Students identify causal mechanisms linking sensory ambiguity, prior experience, and cognitive load to predictable increases in bias strength and decreases in decision accuracy.

CAST Testing Alignment

MULTIPLE CHOICE:

A student with strong prior beliefs about climate change reads an ambiguous news article about temperature data. Based on the cognitive bias model, what is the most likely outcome and why?

CONSTRUCTED RESPONSE:

Using your model, explain why a tired doctor at the end of a 16-hour shift is more likely to misdiagnose a patient than the same doctor at the start of the shift. Reference at least three components from your model.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Every second, your brain processes approximately 11 million bits of sensory information, but your conscious mind can only handle about 50 bits per second. To bridge this enormous gap, the brain uses mental shortcuts called heuristics — fast, efficient rules of thumb that produce good-enough answers most of the time. But these same shortcuts create systematic errors called cognitive biases that affect every human being, regardless of intelligence or education.

The Neuroscience of Heuristics

Daniel Kahneman's Nobel Prize-winning research identified two systems of thinking: System 1 (fast, automatic, intuitive) and System 2 (slow, deliberate, analytical). System 1 handles 95% of daily decisions using heuristics, while System 2 activates for complex problems. Cognitive biases arise when System 1 processes situations that actually require System 2 analysis — the brain takes a shortcut when it should slow down.

Confirmation Bias in Science

Confirmation bias is particularly dangerous in scientific thinking because it causes researchers to unconsciously design experiments that confirm their hypotheses, selectively notice supporting evidence, and discount contradictory results. Studies show that even peer review is affected — reviewers rate papers more favorably when conclusions align with their existing beliefs. This is why double-blind studies and pre-registered hypotheses are critical safeguards.

The Adaptive Value of Bias

Cognitive biases aren't bugs — they're features. In evolutionary terms, the cost of a false positive (seeing a predator that isn't there) is much lower than a false negative (missing a real predator). The brain evolved to be paranoid, pattern-seeking, and quick rather than accurate. In modern environments, these same adaptations cause us to see patterns in random data, jump to conclusions, and resist changing our minds even when evidence demands it.

Cognitive Load and Decision Quality

Research consistently shows that decision quality degrades under high cognitive load. When working memory is taxed — by stress, fatigue, multitasking, or information overload — the brain shifts almost entirely to System 1 processing. This means more reliance on heuristics, stronger biases, and worse decisions. This has critical implications for high-stakes environments like hospitals, courtrooms, and air traffic control.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify sensory input, prior experience, cognitive load, bias strength, and decision accuracy as the key components of the cognitive bias system.

E **Establish Relationships**
Students determine that sensory ambiguity and strong prior experience increase bias strength, that cognitive load amplifies all biases, and that bias strength inversely affects decision accuracy.

V **Visualize & Model**
Students build a computational model showing how the interaction of input clarity, experience, and cognitive load determines the strength of cognitive biases and the accuracy of decisions.

E **Evaluate Outcomes**
Students run clear evidence, confirmation bias, and cognitive overload scenarios to test predictions about when and why the brain produces systematic errors.

R **Revise & Extend**
Students add emotional state, social pressure, or metacognition to explore how real-world factors modify the basic bias model and test bias-reduction strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Today we're investigating something that affects every single one of you. Modeling Cognitive Bias."*

Watch For

Show cover image. Ask: What do you already know about this topic? Quick share-out.

Slide 2: Learning Objectives

3 min

Say: *"Here's what you'll be able to do by the end of today's lesson."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary with quick visual aids.

Slide 3: Big Question

3 min

Say: *"This is our driving question. By the end of class, you'll be able to answer it with evidence from your model."*

Watch For

Quick-write: Students write their initial hypothesis. Save to compare later.

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a computational model to investigate this system. Here are our five steps."*

Watch For

Preview each LEVER step. Emphasize that modeling reveals hidden patterns.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this system? Which ones come from outside, and which respond inside?"*

Watch For

Guide sorting of external vs. internal components. Use think-pair-share for reasoning.

Slide 6: Activity 2: Connections

8 min

Say: *"When Prior Experience is very strong and Sensory Input is weak, what happens to Bias Strength?"*

Watch For

Students draw arrows showing +/- relationships. Discuss feedback loops if present.

Slide 7: Activity 3: Simulation

10 min

Say: *"Now let's test our model. Run each scenario and record what happens."*

Watch For

Students run all scenarios, record predictions vs. observations. Discuss surprises.

Slide 8: Discoveries

5 min

Say: *"So what did our model reveal? Your brain lies because it's optimized for speed, not accuracy...."*

Watch For

Class discussion of discoveries. Compare initial hypotheses to model evidence.

Slide 9: STEM Challenge

12 min

Say: *"Now apply what you've learned. Design a decision-support tool that helps doctors maintain diagnostic accuracy...."*

Watch For

Groups design solutions using model data. Present and evaluate designs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sensory Input, Prior Experience
- **INTERNAL:** Cognitive Load, Bias Strength, Decision Accuracy

Relationships

- **Sensory Input to Bias Strength:** NEGATIVE (-) — When sensory input is clear and unambiguous, the brain has less need for shortcuts, so bias strength decreases. When input is weak or ambiguous, the brain fills gaps with assumptions, increasing bias.
- **Prior Experience to Bias Strength:** POSITIVE (+) — Stronger prior experience creates stronger expectations, which increase bias strength. The brain interprets new information through the lens of what it already believes.
- **Cognitive Load to Bias Strength:** POSITIVE (+) — Higher cognitive load forces the brain to rely more on quick heuristics rather than careful analysis, amplifying all cognitive biases.
- **Bias Strength to Decision Accuracy:** NEGATIVE (-) — As bias strength increases, the gap between perception and reality widens, directly reducing decision accuracy.

Key Discoveries

- Cognitive biases are strongest when sensory input is ambiguous and prior experience is strong — the brain fills in gaps with what it expects to see
- High cognitive load amplifies ALL biases because the brain shifts from careful analysis to quick shortcuts when overwhelmed
- Decision accuracy drops most dramatically when strong prior experience meets ambiguous evidence — expertise can actually increase bias in uncertain situations
- Awareness of bias doesn't eliminate it but can reduce its impact — the model shows that metacognition acts as a moderating variable on bias strength

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Smart people don't have cognitive biases"	Intelligence does not protect against cognitive biases and may actually make some biases worse. Research shows that higher-IQ individuals are better at rationalizing biased conclusions, not at avoiding them. Cognitive biases are features of neural architecture shared by all human brains, not deficits of intelligence.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If you know about a bias, you can just choose not to be biased"	Awareness of bias is necessary but not sufficient to eliminate it. Studies show that simply telling people about biases has minimal effect on their actual decision-making. Biases operate below conscious awareness in System 1 processing. Effective debiasing requires structural interventions — checklists, alternative generation, external review — not just willpower.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Cognitive biases are always bad"	Cognitive biases are the result of heuristics that work correctly the vast majority of the time. Pattern recognition, quick threat detection, and social conformity all serve critical survival functions. Biases only become problematic in specific situations — when evidence is ambiguous, stakes are high, or the environment differs from what the brain evolved for. The goal isn't to eliminate biases but to recognize when they're likely to mislead.